

Ibtesam Hussain

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PROFILE

AI Engineer skilled in **LLM integration, agentic AI systems, and full-stack development/deployment**. Experienced in building **production-ready** AI applications using **LangChain, FastAPI, and cloud platforms**. Passionate about designing intelligent automation pipelines and data-driven solutions at scale.

EXPERIENCE

Aykays Startup — AI Software Engineer Intern (*Onsite, Karachi, Pakistan*)

July 2025 — Sept 2025

- Implemented and fine-tuned **LLMs (GPT-4.1, Mistral 7B)** via LangChain and OpenRouter API, improving understanding and response accuracy.
- Built Flask-based **ML inference APIs** for fine-tuned LLM models and integrated them into deployed AI web apps.
- Delivered **four milestone Full Stack projects** including an **AI Portfolio Generator, Q&A Forum, Product Catalog, and Booking System** using React, Firebase, Node.js, and MongoDB.
- Collaborated via **Slack, Asana, and Git/GitHub** for project management, reviews, and version control.

FAST NUCES — Final Year Project Lead (*Hybrid, Karachi, Pakistan*)

Sept 2025 — May 2026

- Led a **3-member research team**, delivering a **novel hybrid monocular depth estimation system** for **clinical wound imaging**.
- Designed a two-pipeline system** combining Gemini AI-synthesized stereo viewpoints with DA3 Giant monocular depth model, fused via Laplacian pyramid technique.
- Achieved **43.1%** of image pairs reaching **calibrated stereo range** with best **reprojection errors of 5.26 gray levels** across the dataset.

SKILLS

LLM & GenAI: LangChain, OpenAI API, Prompt Engineering, Agentic AI, n8n, CrewAI

Programming Languages: Python, JavaScript, Java, SQL and C/C++.

AI & ML: Tensor Flow, Keras, Pandas, Pytorch, Scikit-learn, NumPy, OpenCV, Deep Learning (CNNs, RNNs, LSTM), Transfer Learning, YOLO, EDA, Data preprocessing, Feature Engineering, MLflow and TensorBoard.

Tools & Deploy: Flask, FastAPI, Streamlit, Node.js, Model Inference APIs, RESTful APIs, Docker, MLOPs and Cloud Deployment (AWS, Azure, Vercel, Render & GCP), Google Colab, Jupyter Notebook, Visual Studio Code, Slack, Jira, Git/Github.

PROJECTS

FYP Research Assistant — (Hybrid RAG System, RAG Evaluation, Python)

- Built Hybrid RAG pipeline over **8 academic papers** combining **BGE-base dense (ChromaDB) & BM25 sparse retrieval**, fused via **RRF** and **re-ranked with cross-encoder**. Achieved **76% on Recall@5, +16pts** over dense-only baseline.
- Designed structure-aware ingestion using **pymupdf4llm & LangChain MarkdownHeaderTextSplitter**; resolved silent **page_num metadata bug** via **per-page** chunking strategy.
- Evaluated** across **3** retrieval configurations (dense-only, sparse-only, hybrid) on a **28-question** test set; ran **RAGAS LLM-as-judge evaluation** achieving Faithfulness **0.82**, Answer Relevancy **0.89**
- Deployed** on **Streamlit** with **@st.cache_resource** model loading (**BGE + cross-encoder loaded once per session**), split retrieval/generation for perceived latency reduction, and streaming LLM output.

JobAgent — (Agentic AI & LLM-Powered Autonomous Job Monitoring System)

- Developed **JobAgent**, an **Agentic AI-powered autonomous job monitoring system** that scrapes company career portals and delivers personalized real-time job alerts via **WhatsApp**.
- Engineered an **end-to-end AI automation pipeline** using **Playwright, BeautifulSoup, SQLite, and LLM APIs** for intelligent job filtering and decision-making.
- Optimized LLM inference cost by processing **only newly discovered jobs**, reducing redundant API calls by approximately **80%**.
- Reduced irrelevant job processing by **70-85%** through **multi-stage filtering, deduplication, and AI-based relevance selection**.
- Built **automated notification workflows** using **Twilio WhatsApp API** for proactive alert delivery and event-driven execution.
- Designed **modular production-style architecture** with logging, scheduling readiness, database separation, and scalable open-source deployment support.

ConsciousDay Agent — (AI WebApp, LLM Integration, Automation)

- Developed an **AI-powered journaling and productivity agent** using **DeepSeek R1 LLM** for personalized user insights.
- Designed a **context-aware system** to **store and retrieve** user history from **SQLite**, enabling more relevant and consistent responses.
- Implemented structured backend workflows to manage user inputs, context retrieval, and AI-generated outputs.
- Deployed the application using **Streamlit** with persistent data handling for improved user experience and retention.

EDUCATION

FAST National University of Computer and Emerging Sciences (NUCES), Karachi

August 2022 — June 2026

Bachelor of Science in Artificial Intelligence | 2x Dean's Honors List

Relevant Courses: Machine Learning, Deep Learning, Artificial Intelligence, Computer Vision, Agentic AI, DSA, OOPs, NLP.